

Experiments 08–11, decomposed against the model

What each notebook stresses, what it actually shows,
and how it should be read in the IWAI paper

Internal note — IWAI 2026 paper

2026-05-15

Abstract

The four most recent notebooks (08 vector-attentional schools, 09 paradigm competition, 10 Fiedler vanguard, 11 endogenous oscillation) each lean on the same multi-agent active-inference model from the IWAI draft (Sec. 4 of `paper/main.tex`) but stress a different axis. This note ties each one back to the model, says exactly which model prediction it tests, reports what we got — including two findings that diverge from the pre-registered hypothesis — and suggests how the experiments should map onto the experimental programme E1–E4 of `paper/sections/05_experiments.tex`. A companion note, `switch_cost_efc.tex`, grounds notebook 11’s attentional switch cost in the policy prior $E(\pi)$ rather than the pragmatic term, and is a prerequisite for §5 below.

1 The model in one paragraph

A population of N agents lives on a graph. Each agent i holds a Gaussian posterior over a world parameter — scalar $\theta \in \mathbb{R}$ in v2 (`src/`) or vector $\theta \in \mathbb{R}^K$ in the notebook-local lift (08–11). One step is: (i) each agent picks an experiment x_i on a discrete grid by softmax over expected information gain $\text{EIG}(x)$ minus a cost $c(x)$; (ii) observes $o_i \sim \mathcal{N}(h_0(x) + h(x)^\top \theta^*, \sigma^2)$; (iii) forms a private posterior via the prior-reverting Gaussian update

$$\Lambda_{\text{carry}} = \rho_{\text{post}} \Lambda + (1 - \rho_{\text{post}}) \Lambda_p, \quad \Lambda_{\text{carry}} \mu_{\text{carry}} = \rho_{\text{post}} \Lambda \mu + (1 - \rho_{\text{post}}) \Lambda_p \mu_p, \quad (1)$$

where (μ_p, Λ_p) is a fixed paradigm prior and the data term $h(x)h(x)^\top/\sigma^2$ is added to Λ_{carry} ; (iv) broadcasts to neighbours and pools precision-weighted under a trust-precision matrix Γ ; (v) updates Γ by Gamma-conjugate inference on neighbour surprisals. Two things matter for what follows: **at $\rho_{\text{post}} = 1$ the update is a ratchet** (precision only grows), so the whole system is a composition of contractions and converges; **at $\rho_{\text{post}} < 1$ unobserved components relax back to μ_p** , which is the only thing in this model that fights consensus.

2 Notebook 08 — vector world and attentional schools

What it tests. The model in §1 promises that the bottleneck on cross-paradigm understanding is *discovery bandwidth through trust*, not a representational limitation. Notebook 08 tries to put that under load. The world is upgraded from scalar θ to $\theta \in \mathbb{R}^K$ with $K = 4$, and the basis $h_k(x) = \exp(-(x - x_k)^2/w^2)$ is a row of Gaussian bumps peaked at $x_k \in \{1, 3, 5, 7\}$ (Fig. 1). The four communities (“schools”) each have their action grid concentrated near one peak, so the Fisher information $h(x)h(x)^\top/\sigma^2$ added by their own experiments is rank-1 and *aligned with one axis*: school s only gains precision along θ_s from its own observations. The other $K - 1$ axes can only be filled in through the trust-weighted pool.

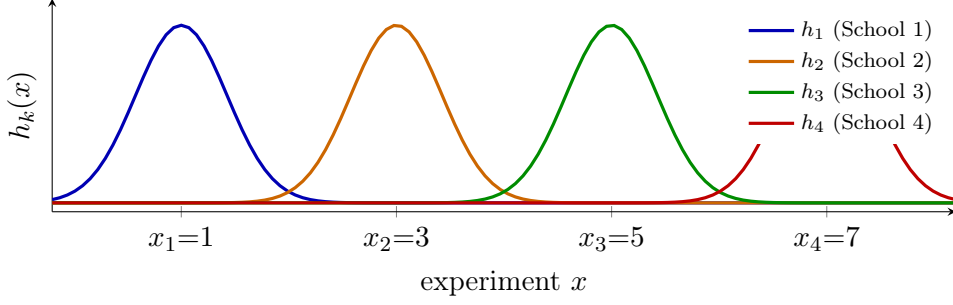


Figure 1: The basis-as-attention. With width $w = 0.6$ the bumps are essentially disjoint: an experiment near x_k provides information about θ_k alone. School s ’s action grid is concentrated near x_s , which makes its private precision update rank-1 along e_s . Cross-school discovery of θ_k for $k \neq s$ has to come from trust.

The protocol. Truth schedule $\theta^*(t) = e_1$ for $t < 60$, then e_3 thereafter — the world shifts from a regime that School 1 can directly probe to one that only School 3 can. The headline plot is the 4×4 grid of per-school mean beliefs $\bar{\mu}_{s,k}(t)$. Diagonal cells (s, s) are *direct learning*; off-diagonals (s, k) for $k \neq s$ are the *import through trust* channel.

Reading. The substantive content is in the off-diagonals. The diagonal cells will rise/fall regardless — a school with direct information on the truth axis tracks it. The interesting question is whether *non-3* schools eventually learn $\theta_3 > 0$ after the shock; if their $(s, 3)$ off-diagonals stay near zero, the population is attentionally locked in the wrong paradigm *even though* every agent’s representation has a θ_3 component. Likewise the inter-school trust matrix at the end of the run is the *discovery topology*: column-3 entries should rise post-shock if the channel has opened.

Where this lands in the model. Notebook 08 is the cleanest operationalisation of the paper’s path-dependence claim (Claim 1 in 04_model.tex, “Trust-driven path-dependence”) without any reversal schedule. The reversal in θ^* is replaced by a *change of probed dimension* — which is what Kuhn’s incommensurability really points at, not a literal change in the laws. Calibration: if cross-school discovery happens, the model’s trust-pooling channel is doing the work the paper claims; if not, the attentional siloing is the binding constraint and the paper should say so explicitly. Either outcome is a *result*, not a failure. The notebook is paper-load-bearing for the substantive claim that “communication structure is the bandwidth of paradigm propagation.”

3 Notebook 09 — paradigm competition under entrenched priors

What it tests. Notebook 08 worked in the *distributional* idiom (full μ, Λ). Notebook 09 specialises it to two *point hypotheses* M_N (Newtonian) and M_R (Relativistic), so the observable becomes the per-agent log Bayes factor. This brings the formal apparatus into direct contact with how working scientists actually adjudicate competing theories. The dimensions are arranged so that θ_N and θ_R disagree on all $K = 4$ axes but with very different magnitudes: $\Delta\theta = (0.5, 1.0, 1.5, 2.5)$. Per-observation expected log-BF contribution scales as $(\Delta\theta_k)^2/(2\sigma^2)$, so at $\sigma = 0.5$ the four communities have predicted discovery rates $(1, 4, 9, 25)$.

The protocol. *Everyone* starts as a Newtonian: $\mu(0) = \theta_N$, $\Lambda(0) = \lambda_0 I$ with $\lambda_0 = 30$ — roughly equivalent to 30 hypothetical observations per dimension of prior commitment. The truth is $\theta^* = \theta_R$, fixed for all t . There is no shock. This is the “Mercury’s perihelion accumulates” story — evidence simply piles up against the entrenched prior, and we ask whether and when each community crosses the decisive-evidence threshold $\log \text{BF} \geq +10$ (Kass & Raftery, 1995).

Reading. Three things to compare against the predicted (1, 4, 9, 25) rates:

1. **Is the conversion order right?** C4 should cross first, then C3, C2, C1 if each community were closed. Departures from this ordering are evidence that trust is doing real work — a slow-evidence community that converts *faster* than its private rate has imported M_R socially.
2. **Is the spread compressed?** Predicted rates span a factor of 25; observed crossing times span far less if the population is well-coupled. The *compression ratio* is a quantitative readout of how much the network shortens the slow communities’ wait.
3. **Does any community fail to cross?** Below some λ_0^* everyone converts; above it the slow- $\Delta\theta$ communities (C1, C2) get *stuck* despite the truth being plainly M_R . That stuck regime is the Bayesian formalisation of paradigm capture — and it is what justifies calling the model an account of *paradigm* dynamics rather than just *belief* dynamics.

Where this lands in the model. Notebook 09 is the *Kuhnian empirical-superiority* face of the paper. The point hypotheses are not a different model — they are a measurement instrument applied to the same Gaussian posterior of §1. The substantive contribution is the λ_0 sweep: it locates the *entrenchment threshold* above which an evidence-rich truth fails to spread, and so makes “paradigm capture” a sharp quantity rather than a vague claim. This notebook is the natural backbone of experiment E4 (paradigm capture) in `paper/sections/05_experiments.tex`.

4 Notebook 10 — the Fiedler bottleneck and the small vanguard

What it tests. Notebook 10 makes the experiment *structural*. It collapses the four-way disagreement of nb09 to a single dimension — $\Delta\theta = (0, 0, 0, 2.5)$ — so only the 4th-band community can detect M_R from M_N first-hand. The mainstream (three communities of 24 agents each, total 72) probe peaks 1–3 where $\theta_N \equiv \theta_R$, so their own observations carry *exactly zero* discriminating signal. The 8-agent *vanguard* (community 4) is the only direct evidence channel. Mainstream conversion can only happen through the trust-weighted pool. The control variable is the normalised-Laplacian Fiedler value λ_2 of the graph; we sweep the inter-community bridge probability and average over seeds to get a continuous λ_2 axis.

The pre-registered hypothesis. Mainstream final log BF is a monotone-increasing function of λ_2 , with a threshold below which the paradigm shift stays trapped in the vanguard.

What we got — and the twist. The strict bottleneck holds: at $\lambda_2 = 0$ the mainstream stays *exactly* at its entrenched start (−93.75, zero variance over 8 seeds). But above zero the relationship is *not* monotone (Fig. 2). Mainstream final log BF rises to a peak of $\approx +106$ at $\lambda_2 \approx 0.056$ and then *declines* to $\approx +58$ at the highest connectivity tested. With 8–11 seeds per level and error bars of ± 12 –25, the non-monotonicity is robust.

Reading the twist. The vanguard is not a λ_2 -independent control — it is the driver. Isolated, an 8-agent vanguard pools precision only with itself, every agent probing dimension 4, so its dim-4 precision compounds without temperance: its log BF goes to +2180. Well-connected, it pools with the (entrenched, dim-4-ignorant) mainstream, which drags its belief back to a sober +71. The non-monotonicity in (a) is then explained by (b): at low-but-nonzero λ_2 the vanguard is sharply overconfident *and* has just enough bridges to pump that overconfidence into the mainstream. At higher λ_2 the same bridges work both ways and moderate the vanguard before its message arrives. The “echo-chamber” framing of λ_2 from the v02 archive does explain the strict $\lambda_2 = 0$ trap, but *not* the shape of the function above zero.

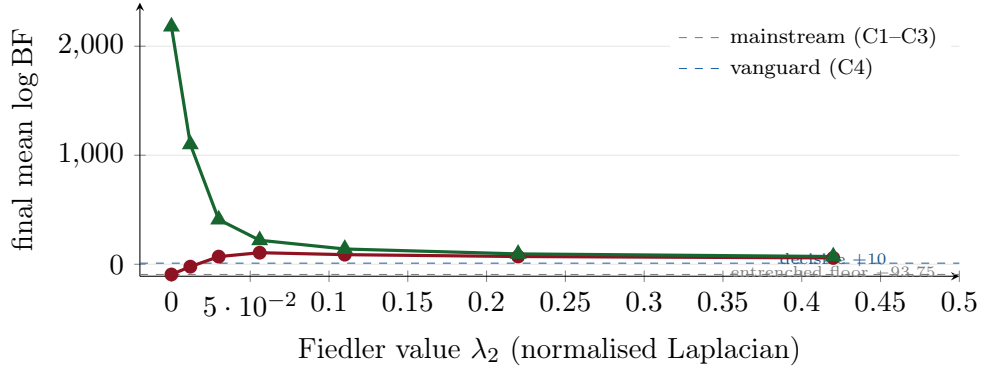


Figure 2: Notebook 10 headline (schematic, values from §8 of nb10). Mainstream final log Bayes factor against structural λ_2 is *non-monotone*: it rises to a peak at $\lambda_2 \approx 0.056$ and then declines. The vanguard’s own log BF is strongly λ_2 -*dependent* — isolated, it explodes to $\approx +2180$; well-connected, it is dragged down to $\approx +71$ by pooling with the entrenched mainstream. The two findings are linked: at low-but-nonzero λ_2 the vanguard is overconfident, and that overconfidence floods the mainstream through the few existing bridges.

Where this lands in the model. The honest take-away is two findings, not one. **(i)** The model exhibits a *strict echo-chamber trap*: with no bridges, the entrenched majority is permanently captured even when decisive evidence is loud one community over. **(ii)** The model exhibits a *vanguard-overconfidence flood*: small communities of like-attending agents amplify their own precision, and below some λ_2 that amplification dominates. The second finding is, on reflection, a *model-level prediction worth making explicit in the paper* — it predicts that paradigm shifts seeded by a small, tightly-knit, evidence-aligned vanguard should overshoot when the bridge to the mainstream is thin, exactly because the vanguard’s epistemic temperature ran away while it was alone. This notebook is the substrate for E2 (topology and community structure), but the contribution should be reframed from “ λ_2 is a threshold” to “ λ_2 controls a competition between two mechanisms.”

5 Notebook 11 — the forgetting bifurcation

The diagnosis. Notebooks 06–10 all produce the same *kind* of object: a trajectory that settles. $\bar{\mu}$ rises and parks; factions merge; the hysteresis loop traces once and dies. That is structural — the model is a gradient flow to consensus. Every component is a contraction: precision-weighted pooling is row-stochastic averaging, the conjugate update at $\rho_{\text{post}} = 1$ is a ratchet, the trust update decays geometrically. By Banach, a composition of contractions has a unique fixed point, and the population finds it.

What this notebook tests. Whether the model can sustain endogenous motion with θ^* held perfectly constant. There are two principled levers (Fig. 3):

- **Posterior forgetting** $\rho_{\text{post}} < 1$ in Eq. (1), so an unobserved component relaxes geometrically toward the paradigm prior μ_p at rate $1 - \rho_{\text{post}}$ while observations pull it toward truth. This is a Hierarchical Gaussian Filter / Kalman-with-reversion construction; it is the only thing in the model that fights consensus. At $\rho_{\text{post}} = 1$ it is bit-identical to the old ratchet (verified: nb06 reproduces exactly).
- **Hysteretic attention.** Each step every agent picks one of $K = 3$ bands by EIG, but with a switching cost c : keep the current band unless another beats it by $> c$. The companion note `switch_cost_efe.tex` grounds this rule in the policy prior $E(\pi)$ of the active-inference

posterior $\sigma(\ln E - \gamma G)$ — it is the MAP action under flat outcome preferences and a persistence-favouring habit $\ln E(a) = -c \cdot \mathbf{1}[a \neq a_{\text{prev}}]$. The dwell time then depends on ρ_{post} , which makes ρ_{post} a genuine oscillator-frequency knob rather than a damping knob.

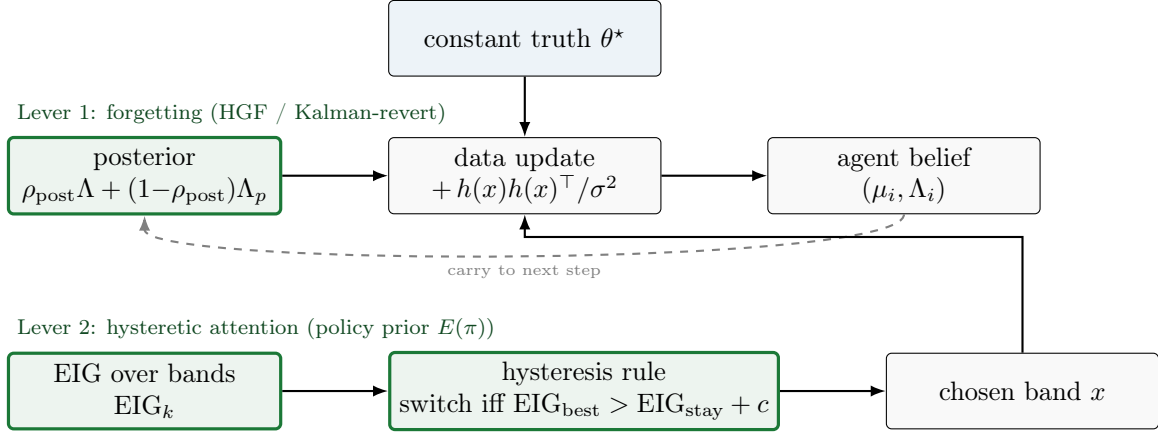


Figure 3: The two principled levers in nb11. Lever 1 fights consensus by relaxing unobserved components back to a fixed paradigm prior μ_p . Lever 2 gives attention inertia, so ρ_{post} controls *when* a neglected band becomes attractive again, and therefore the period of the resulting oscillation. Both levers are independently defensible from the active-inference / predictive-processing literature (see `switch_cost_efe.tex`).

What we got. The model crosses a Hopf-type bifurcation (Fig. 4): below a ρ_{post} threshold the consensus fixed point loses stability and the population sustains an *endogenous oscillation* of belief between the paradigm prior and the evidence, at constant θ^* . Headline numbers at the optimum point ($\rho_{\text{post}} \approx 0.76$, switch cost $c \approx 0.22$, $\rho_{\text{trust}} = 0.95$, coupling $\kappa = 0.15$, constant $\theta^* = (1, 1.5, 2)$, $N = 60$ Watts–Strogatz): per-agent belief amplitude ≈ 0.39 , third-quarter 0.387 vs last-quarter 0.383 (sustained, not transient), largest Lyapunov exponent $\lambda \approx -0.20$ (decisively a stable attractor, not chaos).

Reading the result. Three things to keep separate:

1. **The bifurcation is real and clean.** Both views in §6 of nb11 (the 2-D regime map and the 1-D bifurcation diagram) show a sharp boundary at which a single fixed point opens into an amplitude band. The quarter-by-quarter amplitude check (third quarter 0.387 vs last quarter 0.383) rules out a slowly-decaying transient.
2. **The oscillation is primarily individual; coupling *organises* it.** Per-agent amplitude is ≈ 0.39 ; population-mean amplitude is ≈ 0.03 because heterogeneous phases cancel in the mean. Isolated agents oscillate at ≈ 0.36 ; coupled at ≈ 0.39 . Coupling adds $\sim 8\%$ of amplitude and phase-locks structure — it does not create the motion. The right observable is therefore a per-agent or representative-agent quantity, *not* the population mean.
3. **Chaos is structurally out of reach.** The largest Lyapunov exponent stays at or below the method’s noise floor across the entire (ρ_{post}, c) plane. The mechanism is structural: every component of the model is a contraction; the only non-smoothness is the discrete attention switch, which makes the dynamics piecewise rather than smooth. A piecewise contraction can settle on a periodic orbit but cannot stretch-and-fold (no positive Lyapunov exponent, no sensitive dependence). Genuine chaos would require an *expanding* mechanism — e.g. confirmation bias / theory-laden evidence weighting in which the data update gain

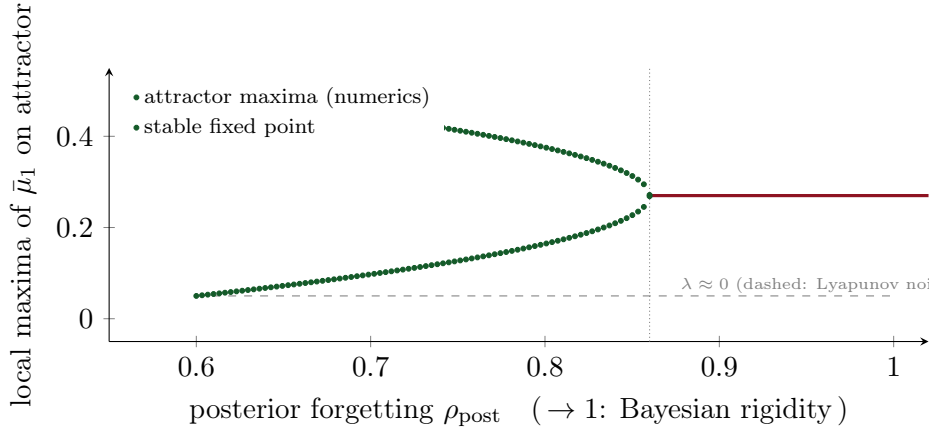


Figure 4: The forgetting bifurcation (schematic, but the qualitative shape and the numerics it summarises are from §6 of nb11). Above the threshold ($\rho_{\text{post}} \gtrsim 0.86$), the consensus fixed point is stable and the attractor is a single point. Below it, the fixed point loses stability and the attractor is a band whose width grows like $\sqrt{\rho^* - \rho_{\text{post}}}$ — the supercritical Hopf signature. The largest Lyapunov exponent stays at the method’s noise floor across the cut (max +0.024): the bifurcation is to a sustained oscillation, *not* to chaos. This is itself a finding: the model class is a piecewise contraction and structurally cannot do chaos.

depends on agreement with the prior. We deliberately do not add it; the negative result is itself a characterisation of the model class.

Where this lands in the model. Notebook 11 is the “result #3” candidate for the IWAI submission — the dissociation from a Bayesian rule-induction baseline. The dissociation axis is ρ_{post} itself: at $\rho_{\text{post}} = 1$ the model is in the convergent class a Bayesian rule-induction account also lives in; below threshold the same model, same constant world sustains an oscillation forever. The defensible paper claim is therefore **“belief forgetting converts a convergence machine into a clean nonlinear oscillator,”** not “the model produces chaos.”

6 How the four notebooks nest

The four experiments are not parallel. They form a stack that progressively stresses the same model along different axes:

The nesting works like this. Notebook 08 establishes that the model can represent attentional siloing at all (and that cross-paradigm discovery has to route through trust). Notebook 09 specialises the same machinery to two competing point hypotheses and finds the entrenchment threshold. Notebook 10 pushes 09 further: it makes the discriminating evidence available only inside a single small community, and asks whether structural connectivity is enough to free the rest — discovering in the process that the answer is non-monotone because the small community itself becomes overconfident. Notebook 11 steps off the convergence axis entirely: it asks, with θ^* constant, whether the model can sustain motion — and finds that one principled change (forgetting) is enough to produce a clean Hopf bifurcation, while the absence of chaos is itself a structural characterisation.

nb	stress axis	world	paper-facing claim
08	<i>representational</i> : scalar $\rightarrow$ vector θ , attention rank-1 in basis	$K = 4$ bumps; truth shock $e_1 \rightarrow e_3$ at $t = 60$	cross-paradigm discovery is trust-bandwidth-limited; trust matrix is the discovery topology
09	<i>decision-theoretic</i> : distributional posterior $\rightarrow$ point Bayes factor, entrenched prior	constant $\theta^* = \theta_R$; entrenchment λ_0 swept	paradigm capture is a threshold in λ_0 above which evidence-rich truth fails to spread
10	<i>structural</i> : discriminating evidence localised to one small community	constant $\theta^* = \theta_R$, $\Delta\theta = (0, 0, 0, 2.5)$; bridge density $\rightarrow \lambda_2$ swept	echo-chamber trap at $\lambda_2=0$ <i>plus</i> a vanguard-overconfidence flood above zero
11	<i>dynamical</i> : ratchet update $\rightarrow$ prior-reverting forgetting; locked $\rightarrow$ hysteretic attention	constant θ^* ; (ρ_{post}, c) swept	Hopf-type bifurcation from convergence to sustained limit cycle (not chaos)

Table 1: The four notebooks against the model. Each one holds the rest of the model fixed and pushes one axis until it breaks something informative.

7 What this means for the IWAI paper

The experimental programme in `paper/sections/05_experiments.tex` declares four experiments E1–E4. Mapping notebooks to E-experiments:

- **E1 (decomposition of three contributors).** Currently declared as a factorial ablation under the reversal schedule $A \rightarrow B \rightarrow A$. None of nb08–11 runs that ablation directly. Notebook 06 (v1 archive) and the v2 reversal experiments do. E1 is *not* satisfied by the current four; it remains the open piece.
- **E2 (topology and community structure).** Satisfied by **nb10**, but the contribution should be reframed: the headline is not a monotone λ_2 threshold but a competition between echo-chamber trapping (at $\lambda_2=0$) and vanguard-overconfidence flooding (low-but-nonzero λ_2). The non-monotonicity is the result, not noise to be averaged out.
- **E3 (experiment-cost regime).** Not directly tested. Note that nb11’s switch cost c is in the *policy prior* slot ($\ln E(\pi)$) rather than the experiment-cost slot $c(x)$ in `04_model.tex` Eq. (2). Both can co-exist; the paper should keep them notationally distinct.
- **E4 (paradigm capture).** Satisfied by **nb09**. The λ_0 sweep (§9 of nb09) is the relaxation-time-divergence experiment E4 asks for. The decisive-evidence threshold reading via $\log \text{BF} \geq +10$ is sharper than the relaxation-time framing and probably the better way to write E4 up.

What needs adding to the paper. Three things, in priority order:

1. **The ρ_{post} knob (Eq. (1)) is missing from `04_model.tex`.** It is in `src/(InferenceConfig.posterior_r` and is the active mechanism in nb11. The model section should add it as the prior-reverting

Kalman update, point at the HGF lineage, and note that $\rho_{\text{post}} = 1$ recovers the ratchet of v2.

2. **The policy prior $E(\pi)$ is missing from Eq. (2) (the policy).** The current $\pi_i(x) \propto \exp(\beta[\text{EIG} - c(x)])$ has no slot for an action-cost / habit term separate from the experiment cost $c(x)$. Notebook 11’s hysteresis cost lives there. The companion note `switch_cost_efe.tex` gives the exact form.
3. **Result #3 should be framed as “forgetting-driven oscillation,” not “chaos.”** The notebook 11 §8 paragraph is the right wording: the model class is a composition of contractions, piecewise; it can produce sustained oscillation but structurally cannot produce chaos. Stating the “no chaos” result as a *characterisation* of the model class is a stronger contribution than waving at chaos and missing.

One honest qualifier. Notebooks 08–11 use a notebook-local lift of the v2 `src/` machinery to vector θ (with $K = 3$ for nb11 and $K = 4$ for nb08–10). The vector inference helpers (`private_update_vec`, `precision_pool_vec`, `surprisal_matrix_vec`) live inside the notebooks, not in `src/`. This is by deliberate project decision (premature abstraction is more expensive than duplication while the experimental programme is still moving). The paper should describe the vector model as the canonical version of §1 and footnote that the scalar implementation is recoverable as the $K = 1$ case.

8 Bottom line

The four notebooks tell a coherent story *about the same model*, not four separate stories. The model is a *contraction-based* multi-agent active-inference system; its dynamical repertoire is *fixed points and (quasi)periodic attractors*. Within that repertoire the model exhibits four distinct paradigm-relevant behaviours: attentional siloing with trust-mediated cross-paradigm discovery (nb08); entrenchment-threshold paradigm capture (nb09); echo-chamber trapping plus a vanguard-overconfidence flood (nb10); forgetting-driven Hopf bifurcation to sustained belief oscillation (nb11). Two of these (nb10’s non-monotonicity, nb11’s no-chaos result) are honest dispatches from a model that didn’t behave the way the pre-registered hypothesis predicted; both are stronger contributions than the original hypothesis would have been. The paper’s job is to name the four behaviours and connect each to the single mechanism that produced it, not to oversell any one of them.

Companion documents

- `notes/switch_cost_efe.tex` — where the nb11 switch cost lives in the active-inference formalism (the policy prior $E(\pi)$, not the pragmatic term).
- `paper/main.tex`, sections `04_model.tex` and `05_experiments.tex` — the model and experimental programme this note interprets the notebooks against.
- `notebooks/0{8,9}*.ipynb`, `notebooks/1{0,1}*.ipynb` — the experiments themselves.
- Memory entries: `project_v2_rewrite_state.md`, `project_vector_world_findings.md` (nb08), `project_notebook_09_paradigm_competition.md`, `project_notebook_10_fiedler_vanguard.md`, `project_notebook_11_endogenous_oscillation.md`, `feedback_honest_structural_findings.md`.